

# Tissue-Specific Separation of Radiation-Responsive DEGs

An Educational Instruction Manual for R-Based Transcriptomic Analysis

Generated by Biomni (Phylo) | 2026-07-09

This manual accompanies an R script that separates differentially expressed genes (DEGs) from an *Arabidopsis thaliana* radiation experiment into tissue-specific response groups. The dataset comes from NASA GeneLab studies OSD-498 and OSD-510, which examined the transcriptional effects of ionizing radiation on whole *Arabidopsis* seedlings. Because the original experiment used whole seedlings (not separated tissues), we use the AtGenExpress developmental expression atlas to infer which tissues each radiation-responsive gene is normally most active in, then group DEGs accordingly.

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# 1. Introduction: The Experiment and the Data

## 1.1 The Biological Question

When plants are exposed to ionizing radiation (such as gamma rays or cosmic radiation), their cells activate DNA damage response pathways, alter cell cycle progression, and modulate stress-responsive gene expression. But different plant tissues and organs may respond differently to radiation. Root cells, which are underground and normally shielded, might mount a different transcriptional response than leaf cells, which are adapted to handle light and oxidative stress. Understanding tissue-specific radiation responses is important for space agriculture, where plants will grow in high-radiation environments beyond Earth's protective magnetic field.

## 1.2 The Dataset

The input file `DEG_OSD498_510_radiation_effect.csv` contains the results of a differential expression analysis performed with DESeq2, a widely used R package for analyzing RNA-seq count data. The data comes from NASA GeneLab studies OSD-498 and OSD-510, which profiled the transcriptome of *Arabidopsis thaliana* seedlings exposed to ionizing radiation.

The file contains 23,573 genes and 8 columns:

| Column         | Description                                      | What It Tells You                                                 |
|----------------|--------------------------------------------------|-------------------------------------------------------------------|
| gene_id        | AGI locus identifier (e.g., AT5G60250)           | Unique gene name in Arabidopsis                                   |
| baseMean       | Average normalized expression across all samples | How highly expressed the gene is overall                          |
| log2FoldChange | Log2 ratio of expression (irradiated / control)  | Direction and magnitude of change; positive = up, negative = down |
| lfcSE          | Standard error of the log2 fold change           | Uncertainty in the fold change estimate                           |
| stat           | Wald test statistic                              | Used to compute the p-value                                       |
| pvalue         | Raw p-value from the Wald test                   | Statistical significance before correction                        |
| padj           | Adjusted p-value (Benjamini-Hochberg FDR)        | Significance after multiple testing correction                    |
| deg_flag       | yes / no flag                                    | Whether the gene is a differentially expressed gene (DEG)         |

Of the 23,573 genes, **6,942 are flagged as DEGs** (`deg_flag = 'yes'`). These are the genes whose expression significantly changed in response to radiation.

### **BEGINNER'S CORNER: What is *Arabidopsis thaliana*?**

*Arabidopsis thaliana* is a small flowering plant widely used as a model organism in plant biology, much like *E. coli* is used in microbiology or *Drosophila* in genetics. It has a small genome (~135 megabases, ~27,000 genes), a short life cycle (~6 weeks), and is easy to grow in the lab. Every gene in *Arabidopsis* has a unique identifier called an AGI code (*Arabidopsis* Genome Initiative), formatted as AT{chromosome}G{number}, e.g., AT5G60250 means chromosome 5, gene 60250.

### **BEGINNER'S CORNER: What is ionizing radiation?**

Ionizing radiation carries enough energy to damage DNA directly, causing single- and double-strand breaks. In space, astronauts and plants are exposed to galactic cosmic rays (high-energy protons and heavy ions) and solar particle events. On Earth, gamma rays from radioactive sources (like Cobalt-60) are used to simulate some aspects of space radiation in lab experiments. Plants have evolved DNA damage response pathways (involving proteins like ATM, ATR, and SOG1) that detect damage and activate repair genes.

## 2. Background: Differential Expression and DESeq2

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### 2.1 What is Differential Expression?

Differential expression analysis compares gene expression levels between two or more conditions (e.g., irradiated vs. control plants) to identify genes that are significantly upregulated (expressed more) or downregulated (expressed less) in one condition relative to the other. The output is a table with a fold change and a p-value for every gene.

### 2.2 Key Concepts in the DESeq2 Output

**Log2 Fold Change (log2FC):** This is the most important column. It tells you the direction and magnitude of expression change. A log2FC of +1 means the gene's expression doubled; -1 means it halved; +2 means it quadrupled. We use log2 (rather than raw fold change) because it makes up- and down-regulation symmetric: +2 and -2 represent the same magnitude of change in opposite directions.

**Adjusted p-value (padj):** When testing thousands of genes simultaneously, some will appear significant by chance alone. The adjusted p-value (using the Benjamini-Hochberg procedure) controls the false discovery rate (FDR). A common threshold is  $\text{padj} < 0.05$ , meaning we expect at most 5% of the genes we call significant to be false positives.

#### BEGINNER'S CORNER: Why log2 instead of raw fold change?

If a gene goes from 100 to 200 reads, that's a 2-fold increase. If it goes from 100 to 50, that's a 0.5-fold change. On a linear scale, 'doubling' (2x) and 'halving' (0.5x) look very different in magnitude. But on a log2 scale:  $\log_2(2) = +1$  and  $\log_2(0.5) = -1$ . Now they're symmetric! This makes it much easier to visualize and compare up- and down-regulation on the same plot (like a volcano plot).

### 2.3 The Challenge: No Tissue Information

The critical challenge with this dataset is that it comes from **whole seedling** experiments. The researchers harvested entire Arabidopsis seedlings (roots, leaves, stems, and all) and extracted RNA from the pooled tissue. This means the DEG file contains no tissue column — we cannot directly tell whether a radiation-responsive gene is primarily active in roots, leaves, flowers, or elsewhere.

To address this, we use an **expression atlas** approach: we look up where each gene is normally most highly expressed (across many tissues, measured in a separate reference dataset), and use that information to infer which tissue's biology the gene is most relevant to. This is an inference, not a direct measurement, but it is a well-established approach in plant genomics.

## 3. The Tissue-Specificity Concept: Expression Atlases and the Tau Index

### 3.1 The AtGenExpress Developmental Atlas

The AtGenExpress project (Schmid et al. 2005, Nature Genetics) is the canonical reference for Arabidopsis tissue-specific gene expression. It profiled gene expression across 79+ diverse tissue samples covering the entire plant life cycle, from seeds to senescing leaves, using the Affymetrix ATH1 microarray. The data is publicly available from the Gene Expression Omnibus (GEO) under accession numbers GSE5629 through GSE5634.

We download six GEO series, each covering a different organ system:

| GEO Accession | Tissue Coverage                      | Samples |
|---------------|--------------------------------------|---------|
| GSE5629       | Seedlings and whole plants           | 24      |
| GSE5630       | Leaves (rosette, cauline, senescing) | 60      |
| GSE5631       | Roots                                | 21      |
| GSE5632       | Flowers and pollen (all stages)      | 66      |
| GSE5633       | Shoots, stems, and shoot apex        | 42      |
| GSE5634       | Siliques and seeds (all stages)      | 24      |

In total, this gives us 237 tissue-specific expression profiles across 20,833 genes. We organize these into a **hierarchical tissue structure**:

| Broad Organ  | Sub-Tissues                                                                                                                    | Samples |
|--------------|--------------------------------------------------------------------------------------------------------------------------------|---------|
| Root         | root                                                                                                                           | 21      |
| Seedling     | seedling_green_parts, whole_plant_pre_bolting                                                                                  | 24      |
| Leaf_Shoot   | cotyledon, hypocotyl, rosette_leaf, cauline_leaf, senescing_leaf, rosette_vegetative, stem, shoot_apex (vegetative/transition) | 78      |
| Flower       | sepal, petal, stamen, carpel, mature_pollen, pedicel, flower_stage_9/10/12/15, inflorescence_apex                              | 90      |
| Seed_Silique | silique_stage_3/4/5, seed_stage_6/7/8/9/10                                                                                     | 24      |

### 3.2 The Tau Tissue-Specificity Index

To quantify how tissue-specific a gene's expression is, we compute the **Tau index**, a standard metric used in genomics. Tau ranges from 0 to 1:

$$\text{Tau} = \text{sum}(1 - x_i / \text{max}(x)) / (n - 1)$$

where:

$x_i$  = expression of the gene in tissue  $i$

$n$  = number of tissues

$\text{max}(x)$  = highest expression across all tissues

**Tau = 0** means the gene is expressed equally in all tissues (ubiquitous/constitutive). **Tau = 1** means the gene is expressed in only one tissue (perfectly tissue-specific). We use a threshold of **Tau  $\geq$  0.6** to classify a gene as tissue-specific; genes below this threshold are labeled 'constitutive' (broadly expressed).

#### INTERMEDIATE NOTE: Why Tau and not just 'highest expression tissue'?

Simply assigning a gene to the tissue where it is most expressed would be misleading for housekeeping genes that are expressed everywhere at similar levels. The Tau index first asks 'is this gene actually tissue-specific?' before making an assignment. A gene with Tau = 0.15 (expressed nearly everywhere) should not be assigned to any single tissue, even if one tissue happens to have marginally higher expression. Tau separates truly tissue-specific genes from broadly expressed ones.

#### ADVANCED NOTE: Limitations of the atlas approach

This method infers tissue relevance from where genes are normally expressed in unstressed plants. It does not measure where the radiation response actually occurs. A gene that is normally root-specific might be induced in leaves by radiation. The atlas tells us the 'default' tissue context of each gene, not the treatment-specific context. For direct tissue-level radiation responses, you would need to irradiate and harvest tissues separately (as in some newer experiments). Also, the ATH1 array covers ~21,000 of ~27,000 Arabidopsis genes; genes not on the array are flagged 'unannotated'.

## 4. Code Walkthrough: Section-by-Section Explanation

This section walks through the R script (`tissue_specific_deg_analysis.R`) block by block. The script is organized into 13 numbered sections. We explain what each section does, why it does it that way, and what the key R functions mean.

### 4.1 Section 0: Package Installation and Loading

The script begins by checking whether all required R packages are installed and loading them. The key packages are:

| Package        | Purpose                                                   |
|----------------|-----------------------------------------------------------|
| GEOquery       | Downloads GEO datasets from NCBI                          |
| ath1121501.db  | Maps ATH1 microarray probe IDs to Arabidopsis gene IDs    |
| AnnotationDbi  | Interface for querying annotation databases               |
| ComplexHeatmap | Creates publication-quality heatmaps                      |
| circlize       | Color mapping for heatmaps                                |
| ggplot2        | General-purpose plotting (bar charts, volcano plots)      |
| UpSetR         | Creates UpSet diagrams for set intersection visualization |
| gridExtra      | Arranges multiple plots on one page                       |

```
# Install missing packages automatically
for (pkg in required_cran) {
  if (!requireNamespace(pkg, quietly = TRUE))
    install.packages(pkg, repos = "https://cloud.r-project.org")
}
```

The `requireNamespace()` function checks if a package is available without loading it. If it returns `FALSE`, we install the package. This makes the script reproducible on any machine.

### 4.2 Section 1: Loading and Validating the DEG Dataset

We read the CSV file and rename the unnamed first and last columns. The first column contains gene IDs; the last contains the yes/no DEG flag.

```
deg <- read.csv(deg_file, header = TRUE, stringsAsFactors = FALSE)
colnames(deg)[1] <- "gene_id"
colnames(deg)[ncol(deg)] <- "deg_flag"
```

We also validate the AGI gene ID format. Nuclear genes follow the pattern `AT{1-5}Gnnnnn`. Genes starting with `ATCG` (chloroplast) or `ATMG` (mitochondria) are organellar genes that won't be in the `AtGenExpress` nuclear atlas — they are flagged as 'unannotated' later.

```
# Check for nuclear AGI IDs (AT1G through AT5G)
valid_nuclear <- grepl("^AT[1-5]G[0-9]{5}$", deg$gene_id)
# Add regulation direction
deg$regulation <- ifelse(deg$deg_flag == "yes",
  ifelse(deg$log2FoldChange > 0, "up", "down"), "non_DEG")
```

### BEGINNER'S CORNER: What does grepl() do?

`grepl()` checks whether each string matches a pattern. The pattern `^AT[1-5]G[0-9]{5}$` means: start (^) with 'AT', then a digit 1-5, then 'G', then exactly 5 digits, then end (\$). This matches valid nuclear gene IDs like AT3G27630 but rejects ATCG00090 (chloroplast) or ATMG01090 (mitochondria).

## 4.3 Sections 2-3: Downloading AtGenExpress and Building Tissue Labels

We download six GEO series using `getGEO()` from the `GEOquery` package. Each series returns an `ExpressionSet` object containing the expression matrix and sample metadata. We then extract tissue labels from the sample metadata and map them to a hierarchical structure of broad organs and sub-tissues.

```
# Download a GEO series
gse <- getGEO("GSE5629", destdir = geodir, getGPL = TRUE)

# Extract tissue from the characteristics column
extracted <- sub(".*Tissue:\\s*", "", vals) # remove "Tissue: " prefix
extracted <- sub("\\s*;.*", "", extracted) # remove anything after ;
```

The `build_tissue_map()` function uses a series of if/else statements with `grepl()` to classify each raw tissue label (e.g., 'flowers stage 12, sepals') into a broad organ ('Flower') and sub-tissue ('sepal'). This is a rule-based mapping because the AtGenExpress tissue labels are free-text descriptions, not controlled vocabulary.

## 4.4 Section 4: Mapping Microarray Probes to Gene IDs

The ATH1 microarray uses probe set IDs (e.g., 244901\_at), not gene names. We use the `ath1121501.db` annotation package to map probes to AGI gene IDs via the TAIR column. Some genes have multiple probes; we keep the one with the highest average expression.

```
# Map probes to AGI gene IDs
agi_map <- AnnotationDbi::select(ath1121501.db,
  keys = probe_ids, columns = c("PROBEID", "TAIR"), keytype = "PROBEID")

# Keep only nuclear AGI IDs
agi_map <- agi_map[grepl("^AT[1-5]G[0-9]{5}$", agi_map$TAIR), ]
```

## 4.5 Section 5: Building Tissue-Level Expression Matrices

We average expression across all biological replicates within each tissue group, producing one expression value per gene per tissue. This is done at both the sub-tissue level (33 tissues) and the broad-organ level (5 organs).



```
# Average expression per sub-tissue
sub_tissue_expr <- sapply(sub_tissues, function(st) {
  samples_st <- all_samples$sample_id[all_samples$sub_tissue == st]
  rowMeans(expr_mapped[, samples_st, drop = FALSE], na.rm = TRUE)
})
```

## 4.6 Section 6: Computing the Tau Index

The `compute_tau()` function applies the Tau formula to each gene (each row of the expression matrix). The `apply()` function with `MARGIN = 1` iterates over rows.

```
compute_tau <- function(expr_matrix) {
  expr_matrix[expr_matrix < 0] <- 0 # clamp negatives
  apply(expr_matrix, 1, function(x) {
    mx <- max(x, na.rm = TRUE)
    if (mx == 0) return(NA)
    n <- sum(!is.na(x))
    sum(1 - (x / mx), na.rm = TRUE) / (n - 1)
  })
}
```

After computing Tau, we classify each gene as 'tissue\_specific' ( $\text{Tau} \geq 0.6$ ), 'constitutive' ( $\text{Tau} < 0.6$ ), or 'unannotated' (not in the atlas). We also identify the predominant tissue for each gene (the tissue with the highest expression).

## 4.7 Section 7: Merging DEG Data with Tissue Annotations

We merge the DESeq2 results with the tissue annotations using `merge()`, which joins two data frames by a shared column (`gene_id`). Genes not in the atlas are flagged as 'unannotated'.

## 4.8 Sections 8-9: Exporting CSVs and Summary Table

We create a hierarchical folder structure and write one CSV per tissue. The summary table aggregates DEG counts, up/down breakdown, and statistics per tissue.

## 4.9 Sections 10-13: Generating Visualizations

Four figures are generated: (1) a stacked bar chart of DEG counts per organ, (2) volcano plots per organ, (3) a heatmap of the top 50 DEGs across all sub-tissues, and (4) an UpSet diagram showing organ expression overlaps. Each uses `ggplot2` or `ComplexHeatmap` with colorblind-friendly palettes.

## 5. Understanding the Outputs: CSVs, Summary Table, and Figures

### 5.1 Hierarchical CSV Files

The script creates a folder called `tissue_specific_degs/` with the following structure:

```
tissue_specific_degs/
├── Flower/
│   ├── Flower_all_DEGs.csv (all flower-specific DEGs)
│   ├── carpel_DEGs.csv (DEGs specific to carpels)
│   ├── petal_DEGs.csv
│   ├── sepal_DEGs.csv
│   ├── stamen_DEGs.csv
│   ├── mature_pollen_DEGs.csv
│   └── ... (one per flower sub-tissue)
├── Leaf_Shoot/
│   ├── Leaf_Shoot_all_DEGs.csv
│   ├── rosette_leaf_DEGs.csv
│   ├── senescing_leaf_DEGs.csv
│   └── ...
├── Root/
│   └── root_DEGs.csv
├── Seed_Silique/
│   └── ... (seed and silique sub-tissues)
├── Seedling/
│   └── ...
├── constitutive_DEGs.csv (broadly expressed DEGs)
├── unannotated_DEGs.csv (organellar / not on ATH1 array)
└── all_degs_with_tissue_annotation.csv (master file: all 23,573 genes)
```

Each CSV contains the original DESeq2 columns plus the tissue annotations:

| Added Column            | Description                                      |
|-------------------------|--------------------------------------------------|
| regulation              | up, down, or non_DEG                             |
| tau_subtissue           | Tau index at sub-tissue level (0-1)              |
| tau_broad               | Tau index at broad-organ level (0-1)             |
| specificity             | tissue_specific, constitutive, or unannotated    |
| predominant_broad_organ | Organ with highest expression (e.g., Flower)     |
| predominant_subtissue   | Sub-tissue with highest expression (e.g., sepal) |

### 5.2 Summary Table (`tissue_deg_summary.csv`)

This table provides a bird's-eye view of the tissue distribution of radiation-responsive DEGs. Here are the key results from this dataset:

| Broad Organ | Total DEGs | Up  | Down | Median log2FC |
|-------------|------------|-----|------|---------------|
| Leaf_Shoot  | 1,399      | 727 | 672  | +0.22         |

| Broad Organ  | Total DEGs | Up    | Down  | Median log2FC |
|--------------|------------|-------|-------|---------------|
| Flower       | 1,192      | 677   | 515   | +0.22         |
| Seed_Silique | 609        | 437   | 172   | +0.24         |
| Root         | 432        | 154   | 278   | -0.25         |
| Seedling     | 72         | 54    | 18    | +0.69         |
| Constitutive | 2,115      | 1,055 | 1,060 | -0.06         |
| Unannotated  | 1,123      | 670   | 453   | +0.21         |
| TOTAL        | 6,942      | 3,719 | 3,223 | +0.12         |

A striking pattern emerges: **root-specific DEGs are predominantly downregulated** (278 down vs 154 up, median log2FC = -0.25), while **seed/silique and seedling DEGs are predominantly upregulated**. This suggests radiation suppresses root gene expression more than it enhances it, while developmental and reproductive tissues show more activation.

### 5.3 Figures

**Figure 1: Bar Chart (fig1\_deg\_counts\_by\_organ.png)** — A stacked bar chart showing the number of upregulated (blue) and downregulated (orange) DEGs in each broad organ. This gives a quick visual overview of which tissues have the most radiation-responsive genes and whether the response is primarily activation or suppression.

**Figure 2: Volcano Plots (fig2\_volcano\_plots\_by\_organ.png)** — Five panels (one per organ), each showing log2FoldChange on the x-axis and  $-\log_{10}(\text{padj})$  on the y-axis. Points above the dashed line are statistically significant. Blue points are upregulated DEGs, orange are downregulated. Grey points are constitutive genes shown for context. Volcano plots help you see both the magnitude and significance of expression changes simultaneously.

**Figure 3: Heatmap (fig3\_heatmap\_top\_degs.png)** — Shows the tissue expression patterns of the 50 most significant DEGs across all 33 sub-tissues. Each row is a gene (z-scored expression), each column is a sub-tissue. The top annotation bar colors sub-tissues by broad organ. The right annotation shows whether each DEG is up- or downregulated in response to radiation. This reveals which tissues the most significant radiation genes normally operate in.

**Figure 4: UpSet Diagram (fig4\_upset\_organ\_overlaps.png)** — Shows how many DEGs are expressed above their median level in each combination of organs. The bar chart on the left shows the total number of DEGs expressed in each organ. The main bar chart shows intersection sizes — how many DEGs are co-expressed across multiple organs. Connected dots indicate which organs are in each intersection. This helps identify DEGs that are broadly active vs. organ-restricted.

## 6. How to Run the Code: Prerequisites and Execution

### 6.1 Prerequisites

You need R (version 4.0 or later) with internet access (for downloading GEO data). The script will automatically install any missing packages.

### 6.2 Step-by-Step Instructions

**Step 1:** Place the DEG CSV file and the R script in the same directory.

```
# Your directory should look like:
# /your_project/
# ■■■■ DEG_OSD498_510_radiation_effect.csv
# ■■■■ tissue_specific_deg_analysis.R
```

**Step 2:** Open R or RStudio and set the working directory to the folder containing your files.

```
setwd("/path/to/your_project")
```

**Step 3:** Run the script.

```
# In RStudio: click "Source" or press Cmd/Ctrl + Shift + S
# In terminal:
Rscript tissue_specific_deg_analysis.R
```

**Step 4:** Wait for the analysis to complete. The GEO download step may take 5-15 minutes depending on your internet speed. The entire analysis typically completes in 10-20 minutes.

**Step 5:** Check the outputs. All files will be created in your working directory:

```
# Output files:
# tissue_specific_degs/ (folder with hierarchical CSVs)
# tissue_deg_summary.csv (summary table)
# fig1_deg_counts_by_organ.png
# fig2_volcano_plots_by_organ.png
# fig3_heatmap_top_degs.png
# fig4_upset_organ_overlaps.png
```

#### BEGINNER'S CORNER: What is RStudio?

RStudio is a free, user-friendly interface for R. Instead of typing commands in a plain terminal, you get a code editor (with syntax highlighting), a console (where R runs), an environment panel (showing your variables), and a plots panel (showing your figures). Download it from [posit.co](https://posit.co). To run a script, open it in RStudio and click the 'Source' button in the top right of the code editor.

### 6.3 Customizing the Analysis

You can modify these parameters near the top of the script:

| Parameter | Default            | Effect of Changing            |
|-----------|--------------------|-------------------------------|
| deg_file  | DEG_OSD498_510_... | Point to a different DEG file |

| Parameter                | Default           | Effect of Changing                                                                   |
|--------------------------|-------------------|--------------------------------------------------------------------------------------|
| TAU_THRESHOLD            | 0.6               | Lower (e.g., 0.5) = more genes called tissue-specific; Higher (e.g., 0.8) = stricter |
| UP_COLOR /<br>DOWN_COLOR | #0072B2 / #D55E00 | Change plot colors                                                                   |

## 7. Interpretation Guide: What Tissue-Specific Radiation Responses Mean

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### 7.1 Key Findings from This Dataset

**Finding 1: Leaf/shoot tissues harbor the most radiation-responsive DEGs (1,399).** This is consistent with leaves being the primary site of photosynthesis and oxidative metabolism. Radiation generates reactive oxygen species (ROS), and leaf tissues have extensive antioxidant systems that may be transcriptionally modulated.

**Finding 2: Root-specific DEGs are predominantly downregulated (64%).** Radiation appears to suppress root gene expression more than it activates it. This could reflect cell cycle arrest (roots are actively growing and dividing, making them vulnerable to DNA damage checkpoints) or a reallocation of resources away from root growth.

**Finding 3: Senescing leaf DEGs are almost entirely upregulated (97%, 426/436).** Radiation may accelerate senescence programs in leaves. Senescence involves the ordered breakdown and remobilization of cellular components, and radiation-induced DNA damage could trigger premature activation of these pathways.

**Finding 4: Seed/silique DEGs are predominantly upregulated (72%).** Reproductive tissues may activate protective pathways to shield developing embryos from radiation damage. This is consistent with the importance of protecting the germline in all organisms.

**Finding 5: 2,115 DEGs are constitutive (broadly expressed).** These are housekeeping-like genes that respond to radiation regardless of tissue context. They likely include core DNA damage response genes (e.g., PARP, BRCA homologs, DNA polymerases) that are needed in every cell type.

### 7.2 Biological Interpretation Framework

When interpreting tissue-specific DEG results, consider these questions:

- **Is the direction consistent with known biology?** If root growth is known to be inhibited by radiation, downregulation of root-specific growth genes makes sense.
- **Are DNA repair genes tissue-specific or constitutive?** Core repair genes should be constitutive (every cell needs to repair DNA). Tissue-specific repair genes might reflect different damage types or repair strategies in different organs.
- **Do stress response genes show tissue specificity?** Leaf-specific stress genes might relate to oxidative stress (from photosynthesis + radiation), while root-specific stress genes might relate to different stress pathways.
- **Are developmental genes affected?** If flower or seed developmental genes are disrupted, this could have implications for plant reproduction in space environments.

**ADVANCED NOTE: Caveats and limitations**

1. The AtGenExpress atlas measures baseline expression in unstressed plants. Radiation may alter tissue-specificity. 2. The ATH1 array covers ~21,000 of ~27,000 genes; ~1,123 DEGs (mostly organellar) could not be annotated. 3. The Tau threshold of 0.6 is a convention, not a biological absolute — results may shift with different thresholds. 4. The original experiment used whole seedlings; true tissue-level radiation responses require tissue-specific irradiation and harvest. 5. Cross-platform differences between microarray (atlas) and RNA-seq (DEG data) may introduce noise.

## 8. Exercises and Extensions for Further Exploration

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This section provides exercises for students to deepen their understanding and suggestions for extending the analysis.

### 8.1 Beginner Exercises

- **Exercise 1:** Open `Root/root_DEGs.csv` in a spreadsheet. Sort by `log2FoldChange`. What are the top 5 upregulated and top 5 downregulated root-specific genes? Look up their functions on TAIR ([www.arabidopsis.org](http://www.arabidopsis.org)).
- **Exercise 2:** Compare the number of DEGs in `Flower/mature_pollen_DEGs.csv` vs `Flower/sepal_DEGs.csv`. Why might pollen have more radiation-responsive genes than sepals?
- **Exercise 3:** Look at the bar chart (Figure 1). Which organ has the most balanced up/down ratio? Which is most skewed? What might this tell you about how that tissue responds to radiation?

### 8.2 Intermediate Exercises

- **Exercise 4:** Modify the `TAU_THRESHOLD` to 0.5 and re-run the script. How does this change the number of tissue-specific vs constitutive DEGs? Is the change dramatic or modest? What does this tell you about the sensitivity of the results to the threshold?
- **Exercise 5:** Extract the constitutive DEGs and perform Gene Ontology enrichment analysis (using R packages like `clusterProfiler` or online tools like DAVID). Are DNA repair genes overrepresented in the constitutive set?
- **Exercise 6:** Create a Venn diagram comparing the DEG sets from two organs (e.g., Root vs Flower). How many DEGs are tissue-specific to both? (Hint: since each gene is assigned to one tissue, you'll need to use the expression threshold approach from the UpSet diagram.)

### 8.3 Advanced Extensions

- **Extension 1:** Integrate Gene Ontology (GO) enrichment analysis for each tissue-specific DEG set. Use the `clusterProfiler` R package with the `Arabidopsis.org.At.tair.db` annotation. This will tell you which biological processes are enriched in each tissue's radiation response.
  - **Extension 2:** Compare these results with other NASA GeneLab radiation studies (OSD-502, OSD-508, OSD-658). Do the same tissue-specific patterns hold across different radiation types (gamma vs HZE vs simulated GCR)?
  - **Extension 3:** Use the newer Arabidopsis single-cell atlas (GSE226097, published in Nature Plants 2025) instead of AtGenExpress for cell-type-level resolution. This would give you much finer granularity (individual cell types rather than bulk tissues).
  - **Extension 4:** Build a gene regulatory network for each tissue-specific DEG set using the SCENIC or WGCNA approach. This would identify transcription factors driving the tissue-specific radiation responses.
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## References:

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